



ER Patient Management System

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Data Structure: Doubly Linked List

Problem Statement

Efficiently manage patients in an ER where critical patients need immediate attention. The system must support adding patients at front (critical), end (normal), specific positions, and removing treated patients.

Proposed Solution

Use Doubly Linked List with bidirectional pointers (next, prev):

- **Node:** patientID, next pointer, prev pointer
- **Operations:** $O(1)$ insertion/deletion at any position
- **Advantage:** Bidirectional traversal and flexible management

Operations & Linked List State

1 `insertAtEnd(101)`

101

2 `insertAtEnd(102)`

101 $\leftrightarrow$ 102

3 `insertAtBeginning(200)`

200 $\leftrightarrow$ 101 $\leftrightarrow$ 102

4 `insertAtPosition(150, 2)`

200 $\leftrightarrow$ 150 $\leftrightarrow$ 101 $\leftrightarrow$ 102

5 `deleteFromBeginning()`

150 $\leftrightarrow$ 101 $\leftrightarrow$ 102
(200 removed)

6 `insertAtEnd(300)`

150 $\leftrightarrow$ 101 $\leftrightarrow$ 102 $\leftrightarrow$ 300